

Task	Method	Year	KG	Technique
KG-enhanced LLM pre-training	ERNIE [35]	2019	E	Integrating KGs into Training Objective
	GLM [102]	2020	C	Integrating KGs into Training Objective
	Ebert [103]	2020	D	Integrating KGs into Training Objective
	KEPLER [40]	2021	E	Integrating KGs into Training Objective
	Deterministic LLM [104]	2022	E	Integrating KGs into Training Objective
	KALA [105]	2022	D	Integrating KGs into Training Objective
	WKLM [106]	2020	E	Integrating KGs into Training Objective
	K-BERT [36]	2020	E + D	Integrating KGs into Language Model Inputs
	CoLAKE [107]	2020	E	Integrating KGs into Language Model Inputs
	ERNIE3.0 [101]	2021	E + D	Integrating KGs into Language Model Inputs
	DkLLM [108]	2022	E	Integrating KGs into Language Model Inputs
	KP-PLM [109]	2022	E	KGs Instruction-tuning
	OntoPrompt [110]	2022	E + D	KGs Instruction-tuning
	ChatKBQA [111]	2023	E	KGs Instruction-tuning
	RoG [112]	2023	E	KGs Instruction-tuning
KG-enhanced LLM inference	KGLM [113]	2019	E	Retrival-augmented knowledge fusion
	REALM [114]	2020	E	Retrival-augmented knowledge fusion
	RAG [92]	2020	E	Retrival-augmented knowledge fusion
	EMAT [115]	2022	E	Retrival-augmented knowledge fusion
	Li et al. [64]	2023	C	KGs Prompting
	Mindmap [65]	2023	E + D	KGs Prompting
	ChatRule [116]	2023	E + D	KGs Prompting
	CoK [117]	2023	E + C + D	KGs Prompting
KG-enhanced LLM interpretability	LAMA [14]	2019	E	KGs for LLM probing
	LPAQA [118]	2020	E	KGs for LLM probing
	Autoprompt [119]	2020	E	KGs for LLM probing
	MedLAMA [120]	2022	D	KGs for LLM probing
	LLM-facteval [121]	2023	E + D	KGs for LLM probing
	KagNet [38]	2019	C	KGs for LLM analysis
	Interpret-lm [122]	2021	E	KGs for LLM analysis
	knowledge-neurons [39]	2021	E	KGs for LLM analysis
	Shaobo et al. [123]	2022	E	KGs for LLM analysis